

Laboratory Assignment 9: Latches and Flip-Flops
ECE 0201: Digital Circuits and Systems
(50 Points)

Name

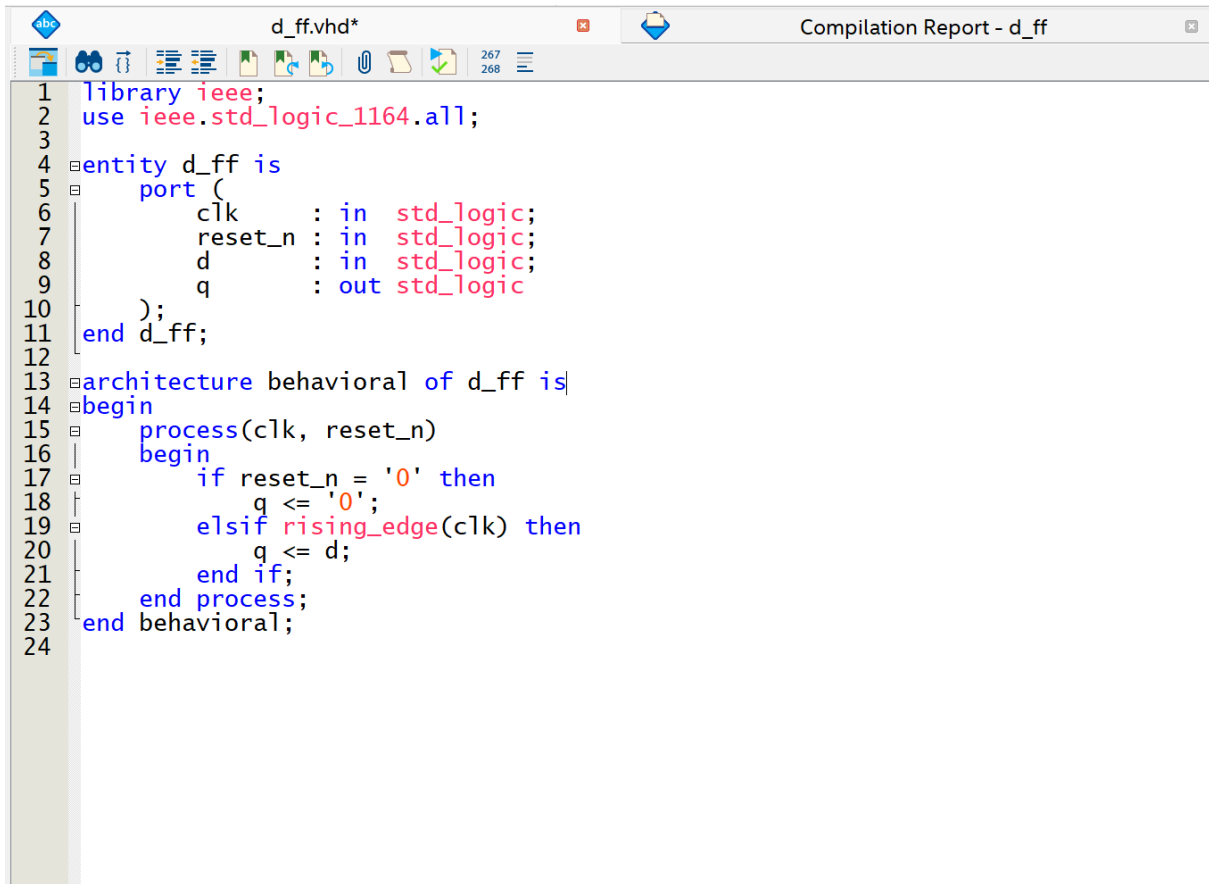
张鹤扬 Stefan

Submission Checklist:

- ☐ Write within boxes, do not move boxes
- ☐ Write your full name in the box above
- ☐ Save this file as a PDF before uploading, keep the number of pages (8) unchanged
- ☐ Note “TO BE CONTINUED” in the answer box if you used the extra pages (7-8)

Part I: D Flip-Flop (15 points)

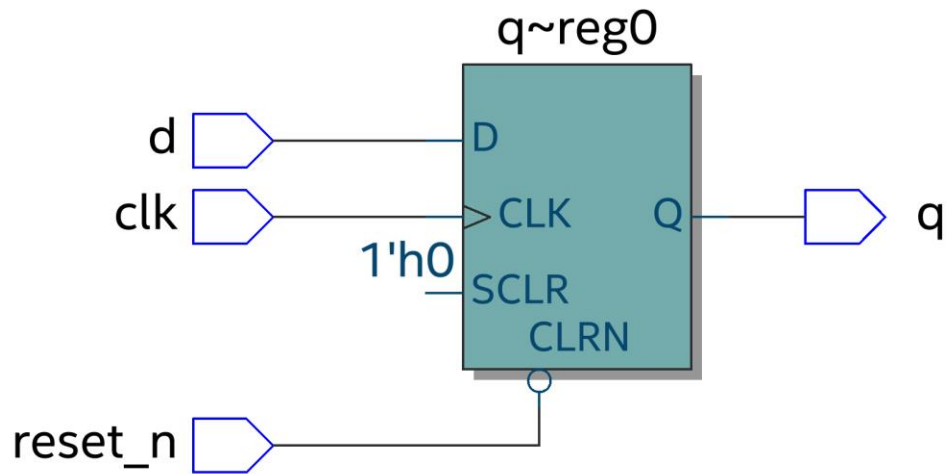
[(A) Insert a screenshot of your VHDL code for Part I.] (3 points)



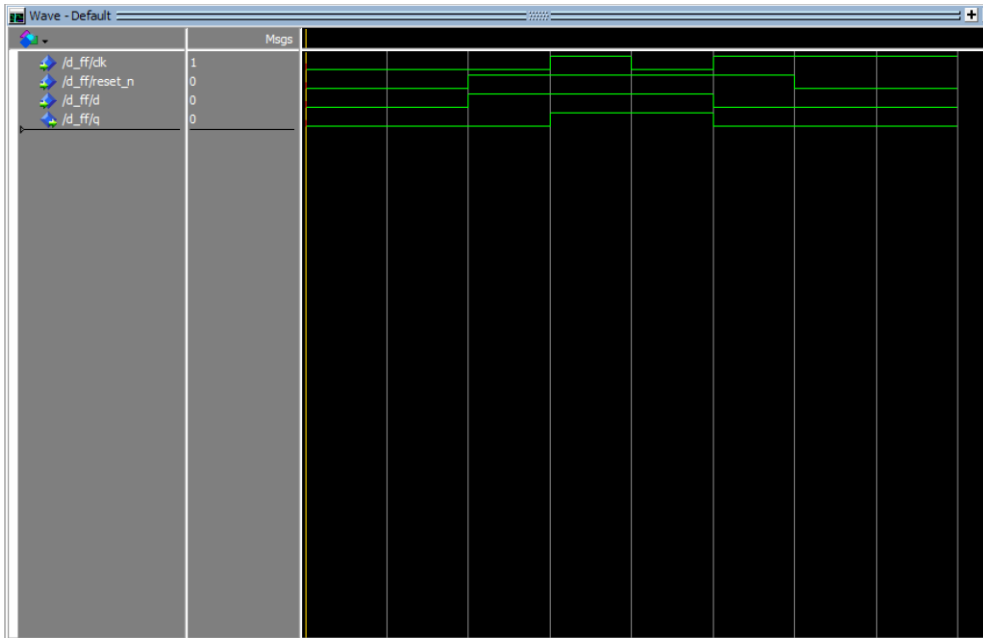
The screenshot shows a VHDL code editor window titled "d_ff.vhd*" with a "Compilation Report - d_ff" tab. The code is as follows:

```
1 library ieee;
2 use ieee.std_logic_1164.all;
3
4 entity d_ff is
5     port (
6         clk      : in  std_logic;
7         reset_n   : in  std_logic;
8         d         : in  std_logic;
9         q         : out std_logic
10    );
11 end d_ff;
12
13 architecture behavioral of d_ff is
14 begin
15     process(clk, reset_n)
16     begin
17         if reset_n = '0' then
18             q <= '0';
19         elsif rising_edge(clk) then
20             q <= d;
21         end if;
22     end process;
23 end behavioral;
```

[(B) Insert a screenshot of your implementation from the RTL Viewer (Tools → Netlist Viewers → RTL Viewer).] (2 points)



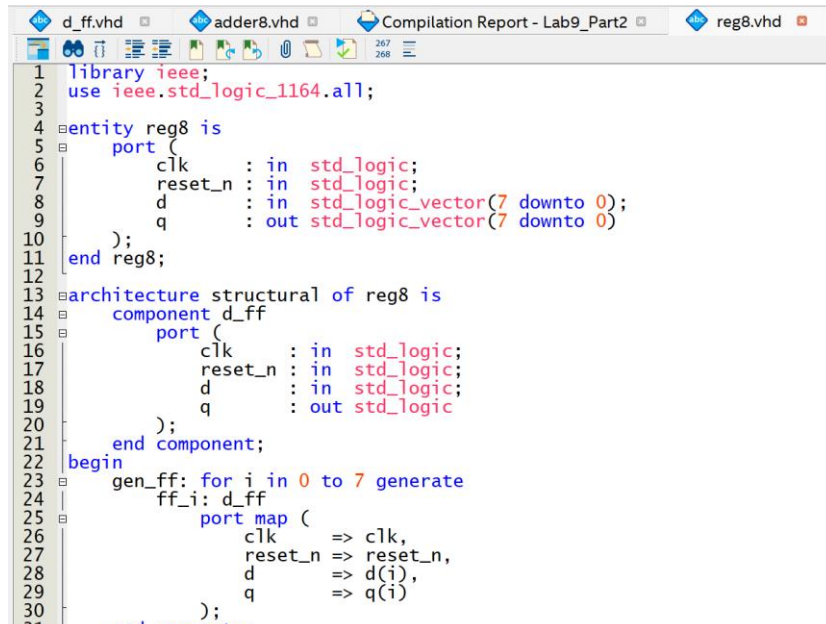
[(C) Insert a screenshot of your functional waveform simulation showing inputs and simulated outputs.] (3 points)



[(D) Demonstrate your circuit to the TA for the first Lab 9 check-off.] (7 points)

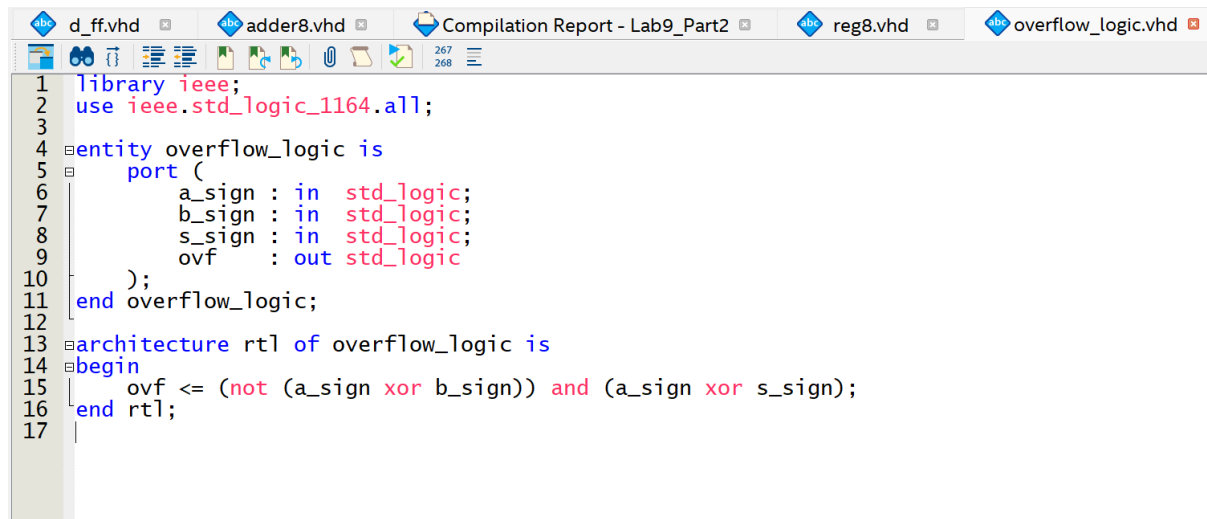
Part II: 8-Bit Accumulator Circuit (35 points)

[(A) Insert a screenshot of your VHDL code for the register component.] (3 points)



```
1 library ieee;
2 use ieee.std_logic_1164.all;
3
4 entity reg8 is
5     port (
6         clk      : in  std_logic;
7         reset_n   : in  std_logic;
8         d         : in  std_logic_vector(7 downto 0);
9         q         : out std_logic_vector(7 downto 0);
10    );
11 end reg8;
12
13 architecture structural of reg8 is
14     component d_ff
15     port (
16         clk      : in  std_logic;
17         reset_n   : in  std_logic;
18         d         : in  std_logic;
19         q         : out std_logic;
20     );
21 end component;
22 begin
23     gen_ff: for i in 0 to 7 generate
24         ff_i: d_ff
25         port map (
26             clk      => clk,
27             reset_n   => reset_n,
28             d         => d(i),
29             q         => q(i)
30         );
31     end generate;
32 end architecture;
```

[(B) Insert a screenshot of your VHDL code for the overflow circuit.] (3 points)



```
1 library ieee;
2 use ieee.std_logic_1164.all;
3
4 entity overflow_logic is
5     port (
6         a_sign : in  std_logic;
7         b_sign : in  std_logic;
8         s_sign : in  std_logic;
9         ovf    : out std_logic;
10    );
11 end overflow_logic;
12
13 architecture rtl of overflow_logic is
14 begin
15     ovf <= (not (a_sign xor b_sign)) and (a_sign xor s_sign);
16 end rtl;
17
```

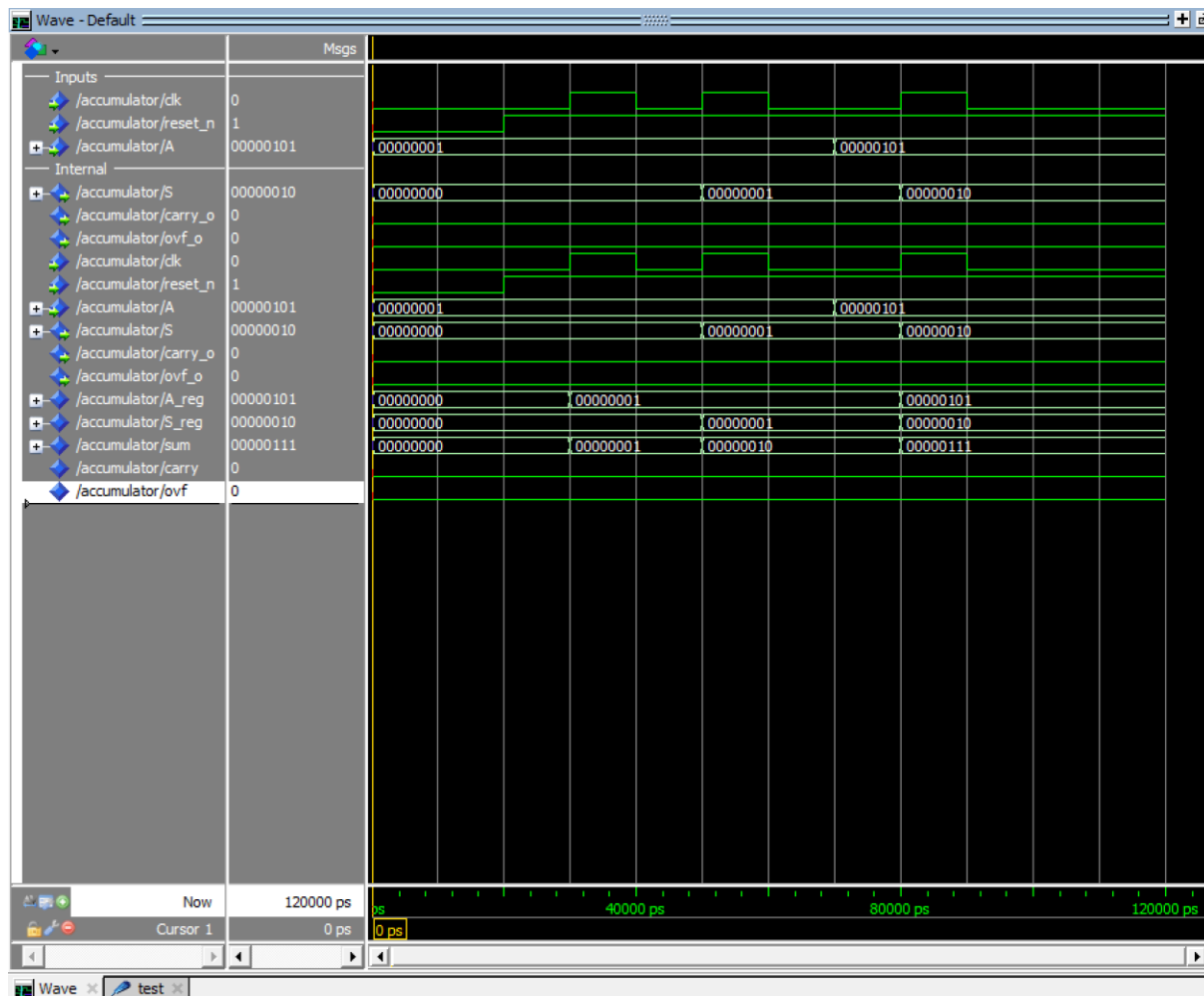
[(C) Insert a screenshot of your top-level VHDL code for the accumulator circuit.] (5 points)

```

d_ff.vhd  adder8.vhd  Compilation Report - Lab9_Part2  reg8.vhd  overflow_logic.vhd
1  library ieee;
2  use ieee.std_logic_1164.all;
3  entity accumulator is
4  port (
5      clk      : in  std_logic;
6      reset_n  : in  std_logic;
7      A        : in  std_logic_vector(7 downto 0);
8      S        : out std_logic_vector(7 downto 0);
9      carry_o  : out std_logic;
10     ovf_o    : out std_logic
11 );
12 end accumulator;
13 architecture structural of accumulator is
14
15     component reg8
16     port (
17         clk      : in  std_logic;
18         reset_n  : in  std_logic;
19         d        : in  std_logic_vector(7 downto 0);
20         q        : out std_logic_vector(7 downto 0)
21     );
22 end component;
23 component adder8
24 port (
25     a      : in  std_logic_vector(7 downto 0);
26     b      : in  std_logic_vector(7 downto 0);
27     cin    : in  std_logic;
28     sum    : out std_logic_vector(7 downto 0);
29     cout   : out std_logic
30 );
31 end component;
32 component overflow_logic
33 port (
34     a_sign : in  std_logic;
35     b_sign : in  std_logic;
36     s_sign : in  std_logic;
37     ovf    : out std_logic
38 );
39 end component;
40 component d_ff
41 port (
42     clk      : in  std_logic;
43     reset_n  : in  std_logic;
44     d        : in  std_logic;
45     q        : out std_logic
46 );
47 end component;
48 signal A_reg, S_reg, sum : std_logic_vector(7 downto 0);
49 signal carry, ovf       : std_logic;
50
51 begin
52     reg_A: reg8
53     port map (
54         clk      => clk,
55         reset_n  => reset_n,
56         d        => A,
57         q        => A_reg
58     );
59     reg_S: reg8
60     port map (
61         clk      => clk,
62         reset_n  => reset_n,
63         d        => sum,
64         q        => S_reg
65     );
66     adder: adder8
67     port map (
68         a      => A_reg,
69         b      => S_reg,
70         cin    => '0',
71         sum    => sum,
72         cout   => carry
73     );
74     ovf_logic: overflow_logic
75     port map (
76         a_sign => A_reg(7),
77         b_sign => S_reg(7),
78         s_sign => sum(7),
79         ovf    => ovf
80     );
81     carry_ff: d_ff
82     port map (
83         clk      => clk,
84         reset_n  => reset_n,
85         d        => carry,
86         q        => carry_o
87     );
88     ovf_ff: d_ff
89     port map (
90         clk      => clk,
91         reset_n  => reset_n,
92         d        => ovf,
93         q        => ovf_o
94     );
95     S <= S_reg;
96 end structural;
97

```

[(D) Insert a screenshot of your ModelSim waveform simulation showing inputs and simulated outputs.] (4 points)



[(E) Demonstrate your circuit to the TA for the second Lab 9 check-off.] (20 points)

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